

“Sounds like a Cheesy Radio Ad”: Using User Perspectives for Enhancing Digital COVID Vaccine Communication Strategies for Public Health Agencies

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ABSTRACT

The World Health Organization (WHO) and other public health agencies have identified vaccine hesitancy as a critical challenge in reducing future cases and deaths from COVID-19. The current study has investigated ways to improve a widely circulated vaccine infographic video by Centers for Disease Control and Prevention. After gathering qualitative feedback on properties of the message that could be improved (from online crowdworkers), we conducted a randomized experiment to investigate different combinations of these attributes. Our results suggest participants were more likely to share the video which was: (1) played more slowly; (2) had a female speaker; (3) did not have background music. The study demonstrates potential of user studies for improving existing communication strategies for encouraging vaccinations and alleviating vaccine hesitancy on social media platforms. Our contribution also includes a repository of messages to encourage vaccination, generated by online crowdworkers, which could be utilized by future studies.

KEYWORDS

vaccine hesitancy, digital communication, social media, public health

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1 INTRODUCTION

COVID Vaccine Hesitancy has been identified as one of the ten major issues that require the World Health Organization’s (WHO) and health partners’ immediate attention. [19]. Alleviating vaccine hesitancy and encouraging public inoculations has been the focus of several avenues of research in the medical and psychology literature. These include using different interviewing techniques to motivate attitudinal change [29], predicting hesitancy through economic indicators [2] and a number of large-scale regional surveys to understand reasons for hesitancy [28]. In the wake of COVID-19 pandemic, a lot of misinformation regarding vaccines have been spreading online [4] further fuelling this hesitancy in the society. This has led to studies which have analyzed content on social media platforms (e.g., Twitter) to understand the rationale behind the anti-vaccine sentiments [18], and deliver targeted digital interventions based on these findings [21].

We have explored the use of user perspectives to suggest improvements to the online communication strategies for vaccination by public health agencies (e.g., World Health Organization, Centers for Disease Control and Prevention, etc). We conducted two studies on Amazon Mechanical Turk (MTurk). Through our first study, we have collected qualitative feedback from the participants about what kind of messages can help in reducing vaccine hesitancy. We use this to design a randomized experiment for our second study to identify the ideal acoustic configuration - namely, gender of speaker, speed of narration, background music, etc for vaccination broadcasts which could be readily shared by social media users. We were guided by the following questions:

- **RQ1:** How do we gain insights into enhancing widely used communication strategies around encouraging COVID vaccination?
- **RQ2:** How do we design messages which can alleviate COVID vaccine hesitancy and have a high probability of being shared online by social media users?

2 RELATED WORK

Existing work on increasing engagement and effectiveness of public health communication on the internet is varied. Researchers have found that social media content with embedded content such as videos or links increase engagement and participation from viewers but may be limited by social media algorithms such as Facebook's [25]. Some others have definitively concluded that posts containing photos are most favorable for high user engagement and posts containing links are most favorable for medium user engagement, [22] for posts relating to public health. Research indicates that the timing of posts is correlated with high engagement and participation. [24]

Specifically, for video content, researchers have found and recommended that shorter videos, median at 6 minutes, find more engagement on MOOC platforms, with a correlation between speaking rate (words per minute) and engagement level. [9] Dialogue-based videos are found to be the most effective in student information retention and video streaming is even believed to alleviate homesickness. [3, 12] Even though post strategies and content might change, public health communicators are recommended to use social media for health-related communication to embrace the social nature of web-based communication, to increase engagement with general audiences, and to improve public health outcomes. [10] Researchers have explored the importance the credibility of the source plays in promoting vaccination, citedred2015 as well as the notion of community building in encouraging vaccination. [6]

Drawing inspiration from this literature, our formative study derived insights from widely used communication strategies for preventing COVID Vaccine Hesitancy (RQ1) and our main deployment study builds upon the insights from formative study to present qualitative results and design recommendations on enhancing a specific kind of message and increasing its shareability on the internet (RQ2).

3 STUDY DESIGN

3.1 Formative Study Survey

We first designed three tasks for Amazon Mechanical Turk [1] as part of a Formative Study, where participants responded to prompts related to COVID-19. The main task allowed all participants to see one form of communication for encouraging vaccination from six - AUD (Audio Action Prompt message), SoCo (Text message about Social Connectedness), IMG (Image informative message), STO (Text based Story prompt), AP (Text based Action Prompt), FAM (Text based Social Connectedness prompt focused on family and community). The prompt asked them to rate the effectiveness of the message shown to them on a scale of 1 ("Not at all effective") to 7 ("Extremely effective"). Then we prompted participants to provide feedback on what they liked about the message, and what they did not like about the message.

In total, we collected 412 valid responses to our online survey study. We recruited 457 adult participants from the US/Canada region, 144 adult participants from UK/Ireland. The participants were majorly White adults (90%), equally distributed in age and gender.

As part of these deployments, we collected 97 valid, free-text feedback on a UNESCO/WHO audio message [27] shown to participants. A grounded theory analysis of these free-text responses was conducted (findings of which are highlighted in the *Results* section). Two coders were involved in this analysis, who developed their own preliminary codebooks, iteratively refined codebooks over several meetings, and applied a final codebook to the data. [7, 16, 23]

3.2 Selection of Video Message for Main Deployment

Our results from the formative study, focused on UNESCO/WHO's audio message, highlighted the shortcomings of the message pointed out by participants. Hence, we wanted to choose a video message that met the following goals:

- Video released by a public health body in North America for the general population because we wanted to focus on building effective communication strategies for COVID-19 vaccines in English-speaking countries with high access to vaccines. The video should also have had a significant number of views.
- Videos containing voice narration with animated information graphics have been recognized as engaging and innovative methods of visual communication using colors and concise information, especially for healthcare [17]. This was the main rationale behind choosing an infographic video. Additionally, to ease the manipulation of the video narration, we did not prefer having a physical and visible narrator in the video.

3.3 Main Deployment Survey

Based on participants' feedback from the formative study, we designed the main deployment with a focus towards the gender of speaker, background music, and playback speed of the informative video as parameters. We constructed a set of 12 (2x2x3) alterations of the video with all combinations of these parameters. The naming convention of each of the arms followed the pattern of [male/female|-|playback speed|-|has background music or not]]. For instance, "female-normal-mus" represents the audio that was spoken by a female speaker at normal speed with background music, and "female-normal-nomus" represents the former audio with the background music removed.

We designed three tasks for Amazon Mechanical Turk to test the effectiveness of a video message encouraging COVID vaccines on the web. The first task addresses the use case where a participant provides feedback on the properties of audio in their assigned video. Participants were randomly assigned to one arm out of the 12 arms. They watched the video and provided free-text feedback specifically about things they liked and things they didn't like about the video they had just watched. They could play the video for unlimited number of times before providing feedback. They could also edit or modify their feedback unlimited number of times.

The second task prompted the participants to rate, on a scale of 1 ("Extremely unlikely") to 7 ("Extremely likely"), if they would share the video on the web (e.g., on their Facebook, Twitter, Instagram

Q103
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On a scale of 1 (extremely unlikely) to 7 (extremely likely), how likely are you to share this video on the web? (for example, on your Facebook, Twitter, Instagram or TikTok account)

1 (extremely unlikely) 2 3 4 5 6 7 (extremely likely)

Build Bank

We would like to understand what you would say to someone who is COVID-19 vaccine hesitant on the web (for example, on your Facebook, Twitter, Instagram, or TikTok account)?

1. Please specify the following when responding to the prompt: What you would say? Please write your message.
2. Please specify the tone of voice you would adopt when conveying this message.

(a) Prompt to provide a shareability rating.
(b) Prompt to build properties of sound message set.

Figure 1: Qualtrics survey deployment study screenshot.

Table 1: Participants' feedback on preferred tones can be divided into four categories: compassionate, assertive, calm, positive.

CATEGORY	Example (s)
Compassionate	"I would say that getting the vaccine isn't only about you, it's about protecting the community around you. About protecting the young and the old, those that are ill. You are apart of the community, so do your part to help your community because we are all in this together. It would stated in a calm and caring voice. One with concern and passion. It would be spoken more slowly."
Assertive	"Please get vaccinated so life can get back to normal. Millions have got it and it has been proven to be effective. The tone of voice I would use while conveying the message would be stern but kind."
Calm	"I know that there's a lot of information going around, so what is it that makes you hesitant about vaccines? Is there anyone in particular you trust most with information like this? You don't have to make a decision about getting vaccinated or not today or by any deadline, so I hope you will consider keeping your options open before you decide against it. I know the news shows a lot of vaccinated people still getting covid these days, but I think it's important to remember that it makes the disease much less severe, so it still protects you and your loved ones. If you have questions about the vaccine itself, I can do my best to answer them or we can look for answers together. I just care a lot about you and don't want to see you and your family get sick. I would use a calm, positive tone of voice to try to convey I cared about them, wanted to know why they think the way they do, and just ask they be open to other possibilities"
Positive	"the vaccine is safe and everyone else is getting it. it is the best thing you can do for you and your family. science is trustworthy and the vaccine works. I would have a positive and persistent attitude and tone of voice."

or TikTok account). The participants were allowed to modify their rating an unlimited number of times (see Figure 1a).

The third task asked the participants, "What would you say to someone who's COVID-19 vaccine hesitant on the web (Facebook, Twitter, Instagram or TikTok account) ?". Participants were also asked to specify the "tone of voice" they would adopt when conveying this message.

3.4 Deployment Data Analysis

We analyzed the participants' feedback responses using the same grounded theory analysis procedures [7, 16, 23] used in our formative study, albeit with a different codebook. Two coders were involved in the iterative analysis procedure. We used the participants' *shareability* ratings to complement our results from the qualitative data analysis.

4 MESSAGE SET FOR ALLEVIATING VACCINE HESITANCY

With the motivation to produce more effective and personalized audio or video messages for tackling COVID vaccine hesitancy, we created a data set which contains a set of curated messages for communication with vaccine hesitant people, along with the description of vocal tone for delivering these messages. We also identified 8 categories of messages (Table 1) (due to space constraints examples of only 4 most relevant are reported). These categories were identified through a qualitative analysis of the data set. All contributors of this bank were participants of the survey, and they were all expected to have seen at least one of the twelve (2x2x3) arms by the time they replied to this prompt. Example responses that were received for each category are displayed in Table 1. Dataset can be found here: <https://publ.cc/NHJwJ>

5 RESULTS

5.1 Formative Study

We discovered that: (i) One of the messages tested contained a *link to family and parents*. Participants who rated FAM highly acknowledged this link as effective, (ii) Participants who provided high-ratings cited *credibility* as an influencing factor for the WHO infographic, and (iii) Participants, in general, liked the idea of including a *sense of community* for AUD and SoCo. These results are promising for RQ1, though a more thorough analysis is required.

5.2 Main Deployment

Data Overview. Results of the experiment are shown in Table 2. The participant pool included a total of 154 participants, of which 52 were females and 102 were males. Our participants were from a wide range of ages - 30% between 20 and 29, 40% between 30 and 39, and about 30% above 40 years of age. Our participants were mostly White (90%) with a few African Americans, Asians and other mixed races.

Speed of narration. Participants allocated to “Female-normal-nomus” or “Female-normal-mus” agreed that the “normal” and “unaltered” speed of the narration was too fast for them to comprehend. Some of the participants appreciated the sound in the original video and thought it was matter-of-fact. However, they expressed a strong preference for *slower* narration speeds. Comparing “Female-slow-nomus” and “Female-normal-nomus”, the average ratings for sharing the video on the web saw a 10% increase in favor of slow speeds, and the increase was 66% when comparing “Female-slow-mus” and “Female-normal-mus” in favor of slow speeds, for the same number of participants in each ($n = 40$).

Presence or Absence of Background Music. Participants observed that the presence of background music in a COVID vaccine video made it sound like an “advertisement”, a “sales advertisement” or a “radio ad”. They conveyed that the presence of background music can lead the listeners to misinterpret the gravity of the content in the video, which as they wrote, “can take the listener’s attention away from the main message of the video.” A few of our participants ($n < 10$) also noted that the music sounded “cheesy” and “annoying”. This caused them to disengage with the content and they were not able to grasp all the pointers of the video in an

efficient manner. Interestingly, our results show that participants seem to react differently to background music when speakers of different sex narrate. When we compare “Female-slow-mus” and “Female-slow-nomus”, we see a 10.7% increase in favor of no background music. Similarly, this behavior also exists in the [female x fast speed] arm pair, in which mean shareability rating increases by 15.9% after removal of the background music. However, when we compare “Male-slow-mus” and “Male-slow-nomus”, there is a 11.1% decrease in mean ratings. The same behavior is also observed in the [male x fast speed] arm pair with a 19.0% decrease.

Voice of narration. Our study highlighted a consensus amongst the participants over “clarity” and “ease of understanding” when narration was by an automated text to speech software. However, participants noted that text to speech software voices sound “robotic and monotonous” and were not engaging. They expressed strong preferences for narrators that are *real human beings with “accents”* so that the content can be more personalized, and pleasant to listen to.

Various more examples for each of the above are included in Table 3.

6 DISCUSSION AND FUTURE WORK

One of the most compelling results of our study is the reproduction of earlier findings that suggest that accented health communications and principles of narrative theory can affect people’s response to health communication, and reduce risk-taking behavior [5, 14, 26]. Our participants have expressed strong preferences for actual, human narrators - e.g., having accented English voices - for greater engagement and better comprehension of the content in online videos. Along with this, they also preferred to have no background music to maintain the gravity around COVID vaccinations and to convey a more focused message. Our results indicate that our current understanding of acceptable speed of narration needs to be modified in order to amplify the effectiveness of video content for vaccines. Previous literature shows that faster speed of online educational content can improve learning outcomes [13], while we show that that may not necessarily be true in the context of public health. These insights are indicators of online content design guidelines that can help alleviate COVID vaccine hesitancy and increase probability of this content being shared online by social media users (RQ2).

As part of this study, we have created a data set for the community to use towards the goal of designing more effective digital communication for COVID-19 vaccines. From our analysis, it seems that calm and compassionate tone of voices are the most effective when talking to vaccine hesitant people, though more information may need to be provided by a participant (e.g., reasons behind adopting a particular tone) for real-world applications to answer RQ2. We faced several challenges with audio recording on Qualtrics software; our next steps include building an audio data set of participants’ voice recording responses to the prompt in Fig 1b.

As this work is primarily about enhancing digital communication pertaining to COVID vaccines, our analysis is intended to be illustrative rather than comprehensive. Our main study is focused on the North American English-speaking population, thus restricting the generalizability of our results. Future work can explore speech

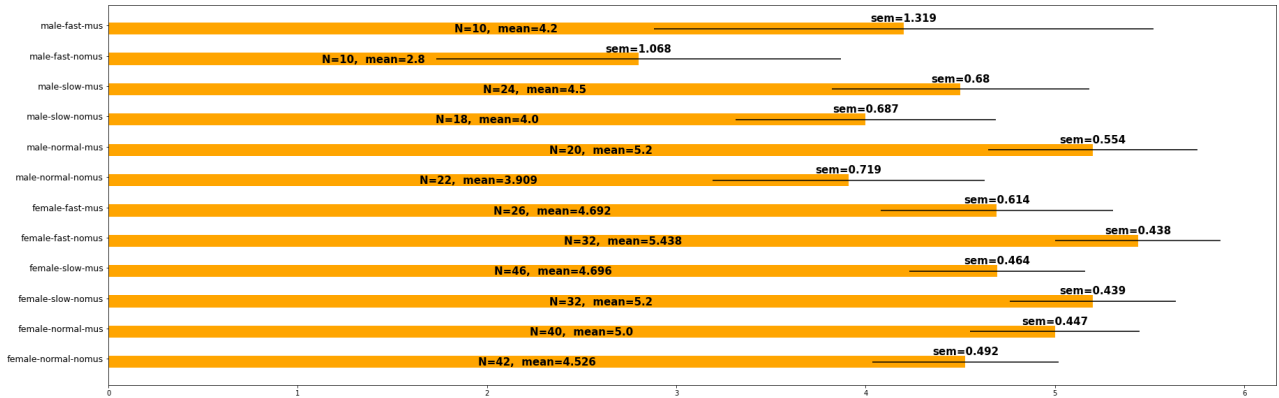


Figure 2: The mean sharability ratings for each arm.

Table 2: Mean, standard deviation, standard error mean and sample size of rewards.

Group	Mean	SD	SEM	N
Female Speaker	4.94	2.04	0.33	74
Male Speaker	4.57	2.24	0.34	80
No Background Music	4.53	2.12	0.31	82
Background Music	4.21	2.32	0.38	72
Normal Speed	4.64	2.15	0.43	50
Slow Speed	4.66	2.18	0.40	60
Fast Speed	5.04	2.05	0.43	44

analysis for different communities with different regional accents. A wider selection of experimental factors such as participant proficiency with social media platforms, randomizing the content of the chosen video could also yield more meaningful understanding and determine which audio properties are more effective than others to increase online shareability.

7 CONCLUSION

In this work, we specify design methodologies to enhance messages aimed at alleviating COVID Vaccine Hesitancy and we provide recommendations on improving the acoustic properties of a popular video encouraging COVID vaccinations, shared by Center for Disease Control and Prevention (CDC). We have presented promising results in increasing shareability of COVID Vaccine videos on the web or social media with these guidelines. Our work also builds a novel dataset with personalized messages and desired tonal properties from online crowdworkers, such as Amazon MTurk. Although a more thorough analysis and varied experimental factors are needed, we are confident that this understanding will aid public health bodies and government agencies to design more effective online COVID vaccination campaigns.

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Table 3: Examples of participants' feedbacks on the different arms.

Categories of Sound Properties					
Speed of Narration		Presence/Absence of Background Music		Voice of Narration	
Fast	"Fast voice", "spoke too fast for my taste", "too fast to understand",	Presence of background music	"elevator type music...really took away from the message", "sounds like advertisement for credit unions"	Original video voice	"Narrator sounds calm and warm", "didn't like that it was spoken by a white person and not someone with an accent"
Normal	"video very fast", "I didn't like that it was too fast", "narrator should slow down"				
Slow	"voice wasn't too fast that I wasn't able to understand", "very nice narration"	Absence of background music	"sounded serious and conveyed the message about vaccine", "clear voice and easy to understand"	Altered Videos with Text-to-speech software	"Robotic and monotonous voice", "voice is clear and easy to understand", "this tone and voice is hard to engage with", "this speech was very clear", "this voice is not friendly and warm"

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Table 4: Findings from formative study.

Findings from Formative Study			
	Findings	Examples	Implications for current deployment
Voice Characteristics	Participants expressed strong preferences for "soothing" and "friendly" voices. Some participants also noted that "monotonous" voices are harder to engage with	"it could be improved by having a more soothing voice so that it does not sound so much like an advertisement but more like a friendly, inviting conversation." "change the presenter, the voice is bland and won't capture the attention of the younger generation."	There remains some ambiguity in the choice of voice for narrations in scenarios centered around online information, and public health. [11, 20] Hence, we decided to test "Male" and "Female" voice narrations in our current deployment.
Background Music	Participant feedback highlighted that the presence of background music in the audio clip sounded like a "radio ad", or a "sales advert" rather than public health information. A few participants found that "off-putting" and "annoying". However, some participants also stressed that the presence of background music to be "engaging".	"It has the sound of a cheesy radio ad though, and I find that slightly off-putting - something about the music I think." "I loved the background music and overall tone of the message. It was engaging and made the message memorable."	Given the participants' divided sentiments about the presence of background music, we designed two variations with respect to background music in our deployments - video with and without background music to derive our insights.
Speed of Voice Narration	Participants commented on the speed of the audio message as being "too fast", and the reason behind the message sounding like a "sales advertisement". This increased the cognitive burden of the participant to remember key takeaways from the message. However, some participants found the speed to be "good" and "perfect" - that made it difficult to conclude if the speed of voice narration was right or not.	"The fast talking tone of the speaker makes it sound slightly like a sales advert which could be improved." "The speaker is very friendly the narration is just perfect! The length of the message is also good (30 seconds)"	Studies conclude that people can speak an average of 100 words per minute (wpm). [15] Based on participant feedback and the literature, we chose to design three variations of the chosen video [8] - fast voice narration (~170 words per minute), keeping the original speed of voice narration (~150 words per minute) and slow voice narration (~100 words per minute).

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